

**Client name:** None    **Sex:** Male

**Date of Birth (age):** test (41Y)    **EEG acq. date:** 2019-05-06 14:00:30.520000+00:00

## Brain Function Connectivity

Functional connectivity of the brain is closely related to brain plasticity and resilience. Brain plasticity refers to the brain's ability to change structurally and functionally in response to environmental changes, learning, experience, or injury. This process of neural network modulation and reorganization can be described through functional connectivity measured using quantitative EEG analysis.

In general, a brain that has undergone healthy development maintains balance within and between functional regions. However, the brain may reinforce connectivity in specific regions when adapting to environmental demands or when structural or functional modifications are required. This adaptive capability is an inherent characteristic of the human brain.

Individuals may show variability in functional connectivity patterns depending on long-term learning, professional training, and environmental exposure. For example, individuals engaged in analytical or logical tasks may demonstrate relatively stronger connectivity in left-hemispheric networks, whereas individuals engaged in artistic or social activities may show relatively stronger right-hemispheric connectivity.

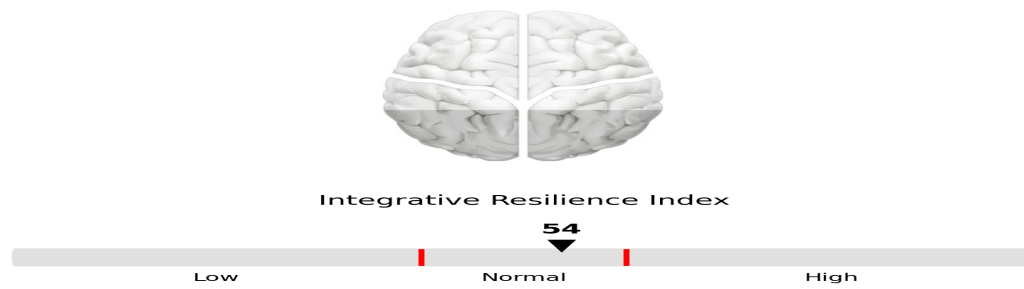
Excessively increased connectivity may be associated with inefficient information processing, excessive mental effort, or reduced flexibility of neural networks. Conversely, insufficient connectivity may be associated with reduced cognitive efficiency and difficulties in coordination between brain regions.

Brain functional connectivity analysis in this report is presented in three categories: integrated functional connectivity of the whole brain, hemispheric connectivity within and between the left and right hemispheres, and local connectivity within frontal and posterior brain regions. These indices provide information related to network balance and functional organization of the brain.

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## 1 Integrated Brain Functional Connectivity



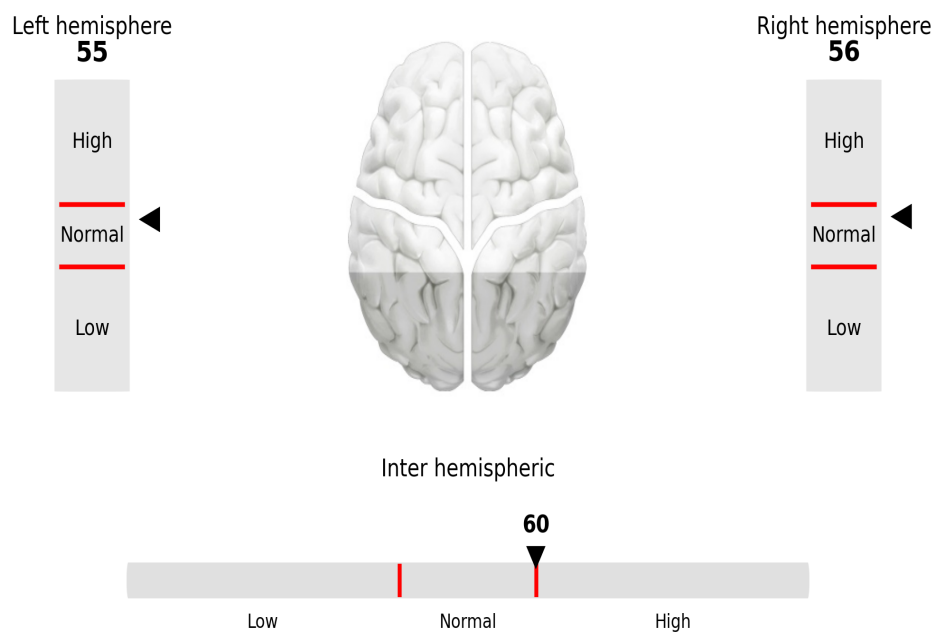
### Analytics results

Both the left hemisphere (55), and right hemisphere (56) have normal connectivity.

- The connectivity between the right hemisphere and left hemisphere is normal (59).
- Left frontal brain functional connectivity is normal (41).
- Right frontal brain functional connectivity is normal (43).
- Left posterior brain functional connectivity is high (70).
- Right posterior brain functional connectivity is high (63).

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## 2 Hemispheric Connectivity



### Analytics results

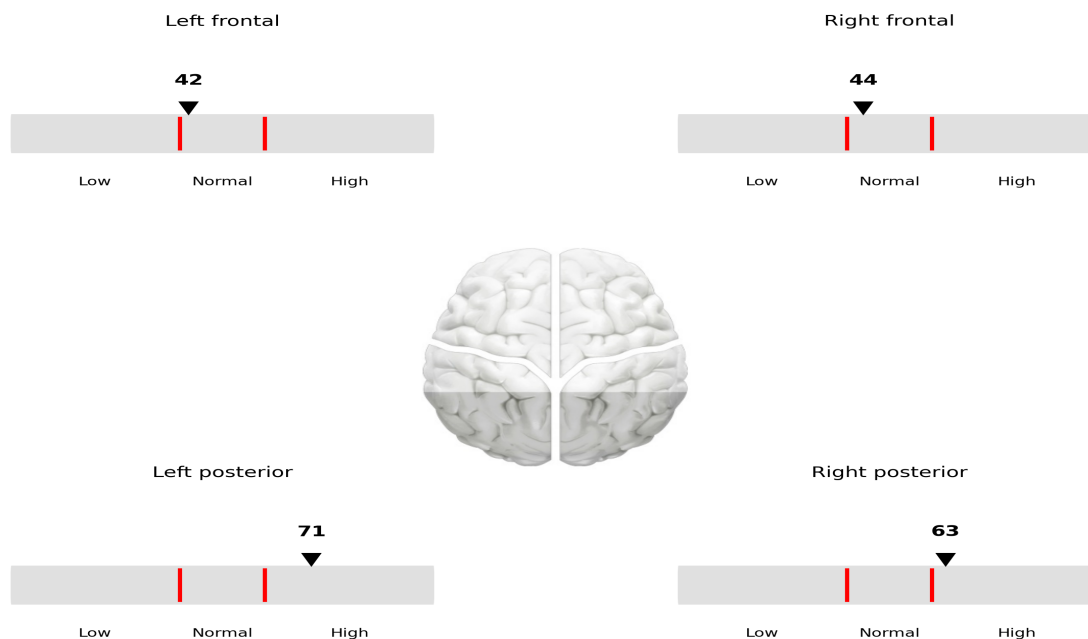
Both the left hemisphere (55), and right hemisphere (56) have normal connectivity.

The connectivity between the right hemisphere and left hemisphere is normal (59).

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### 3 Local Connectivity



#### Analytics results

- Left frontal brain functional connectivity is normal (41).
- Right frontal brain functional connectivity is normal (43).
- Left posterior brain functional connectivity is high (70).
- Right posterior brain functional connectivity is high (63).

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*This result is based on AI-assisted quantitative EEG analysis using default normative values and may differ from actual clinical conditions. This report is not a diagnosis.*